

Human, Synthetic, or Both? Evaluating Training Data Choices for Small Language Models

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Abstract

Recent progress in large language models (LLMs) has been driven by access to large-scale, high-quality human-written text. As data collection becomes increasingly costly, synthetic text generated by powerful LLMs has emerged as a potential alternative or supplement. This work investigates whether small language models can be effectively trained using synthetic data under constrained compute and data budgets. I fine-tune a 70M-parameter Pythia language model on three types of datasets: human-written Wikipedia text, synthetic text generated by Flan-T5 and Gemma-2B, and mixtures of human and synthetic data. Models are evaluated using language modeling perplexity on held-out Wikipedia text, reasoning performance on ARC-Easy and HellaSwag, and qualitative analysis of text continuation. Results show that models trained on human-written data achieve the lowest perplexity, reflecting alignment with the model’s pretraining distribution. However, synthetic data, particularly from Gemma, yields substantially improved reasoning performance despite higher perplexity. Introducing a small proportion of human-written data (20%) into synthetic training significantly reduces perplexity while preserving reasoning gains. These findings highlight that the quality of synthetic data and the choice of generator strongly influence downstream performance, and that hybrid training regimes can offer an effective trade-off between language fidelity and reasoning ability. Overall, this study demonstrates that carefully constructed synthetic data can serve as a viable and competitive training signal for small language models.

1. Introduction

Recent progress in large language models (LLMs) has been driven primarily by increases in data scale, model size, and computational capacity. However, as models and datasets continue to expand, so do concerns around data quality, representational diversity, and the long-term sustainability of

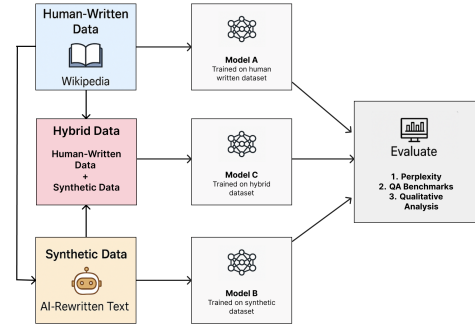


Figure 1. Human-written (Wikipedia) and AI-generated synthetic datasets are preprocessed and also combined to train identical small-scale language models to compare the impact of data source on model performance.

relying exclusively on human-written data. Most publicly available text data is either redundant, noisy, or saturated, limiting its ability to support continued model scaling. To address this bottleneck, researchers have turned to synthetic data, text generated by LLMs themselves as a new frontier in efficient and controllable pretraining.

Synthetic data generation allows for targeted, diverse, and ethically safer datasets that can complement or replace traditional datasets. In recent years, multiple studies have explored how synthetic text can enhance model learning, reasoning, and transferability. Model performance improves significantly when synthetic data is generated with high topical diversity rather than uniform rewriting [5]. Diversity within synthetic corpora mitigates overfitting and fosters better generalization on downstream tasks. Similarly, it has been shown that fine-tuning on carefully crafted synthetic data can yield performance comparable to large-scale human datasets, enabling small models to adapt to specialized tasks with minimal labeled data [2].

A related line of work concluded that hybrid training yields superior robustness and reasoning performance across benchmarks such as MMLU and HellaSwag [8]. Their results indicate that synthetic and human data play

complementary roles: synthetic text improves coverage and uniformity, while human text anchors linguistic and cultural authenticity. In parallel, synthetic datasets not only maintain task performance but also offer privacy-preserving alternatives to using real human data, especially in sensitive domains like healthcare and finance [7].

Moreover, there is a growing use of synthetic data for retrieval and reasoning augmentation [1]. Synthetic text could be used to train specialized retrievers that outperform models trained on natural datasets in open-domain QA and summarization. Syntriever is a framework in which synthetic query–document pairs generated by LLMs significantly boosted retrieval accuracy across diverse tasks [4]. Together, these findings support a broader hypothesis: synthetic data does not merely replicate human text—it actively expands the conceptual and contextual boundaries of training corpora.

Despite these promising results, an open question remains: how effective is synthetic data for small-scale models under constrained compute budgets? Much of the existing literature focuses on billion-parameter LLMs, leaving the small-model regime underexplored. Understanding this space is particularly relevant for research environments with limited resources, where efficient pretraining could enable lightweight models to achieve competitive language understanding and reasoning capabilities.

This project aims to investigate this gap by conducting a controlled comparison between human-written and synthetic datasets in the training of small language models (less than 100M parameters). By pretraining identical architectures on human versus synthetic versions of the same dataset (e.g., Wikipedia), this study quantifies how the data source affects model perplexity, reasoning accuracy, and generalization. Building on prior work, this study seeks to evaluate whether synthetic data can serve as an efficient, scalable alternative to human data, particularly for researchers and practitioners operating in low-resource environments.

2. Background

Modern large language models (LLMs) such as GPT, LLaMA, and Mistral rely heavily on the scale and quality of their training data. However, as publicly available human-written data becomes saturated and harder to curate, researchers are turning to synthetic data, AI-generated text produced by stronger models, to supplement or replace traditional datasets. Synthetic data offers three main advantages: scalability, controllability, and domain adaptability.

Recent work has shown that combining synthetic and human data can substantially improve model robustness and contextual understanding. Zhezherau and Yanockin (2024) demonstrated in Hybrid Training Approaches for LLMs that models fine-tuned on mixed real and synthetic data out-

perform those trained solely on either source, especially in domain-specific applications such as mental health counseling [8]. Their findings emphasize that real-world data, though authentic, is often noisy, incomplete, or narrow in scope, while synthetic data can fill gaps by creating diverse and edge-case scenarios that improve a model’s adaptability.

Similarly, Yu et al. (2023) in Training Private and Efficient Language Models with Synthetic Data from LLMs explored how synthetic data can enable lightweight models to achieve near state-of-the-art performance with better privacy guarantees [7]. Their work at Microsoft Research showed that small models trained on synthetic text generated by larger, differentially private LLMs retained over 97% of the performance of non-private counterparts, demonstrating that synthetic datasets can act as a strong medium for knowledge transfer. This approach reduces data privacy risks and training costs while preserving linguistic and contextual richness.

Complementary research by Kim and Baek (2025) introduced Syntriever, a framework that uses synthetic data to train dense retrievers, showing that synthetic query–document pairs generated by LLMs significantly improve retrieval accuracy across diverse benchmarks [4]. Their results confirm that LLM-generated synthetic data can encode meaningful relationships between concepts, aiding smaller models in reasoning and generalization.

A paper by Xu Guo and Yiqiang Chen (2024) surveys how large language models (LLMs) generate synthetic data for low-resource domains. It discusses prompt engineering, data quality evaluation, and ethical concerns, emphasizing the potential of LLMs to bridge data scarcity while highlighting issues like bias and hallucination [3].

The Blogsynth project demonstrates that synthetic data generated from strong teacher models can be competitive with real data for training smaller language models, while also offering greater control over content diversity and structure [6]. However, this line of work also emphasizes that the effectiveness of synthetic data depends critically on the quality of the generator model and the distributional alignment between synthetic and target data.

Collectively, these studies suggest a growing consensus: synthetic data is not merely a substitute but an enhancer of human data, capable of increasing diversity, domain coverage, and privacy in LLM training. However, few works have systematically evaluated this effect for small-scale, low-resource models. This project builds on these insights by experimentally comparing how synthetic versus human-written corpora affect small LLMs’ performance in language modeling and reasoning tasks.

3. Method

The goal of this project is to compare the performance of small-scale language models trained on human-written versus synthetic (AI-generated) data using a controlled experimental design. The process consists of five main phases: human-written data collection, synthetic data generation, human-written data + synthetic data mixing, model training, and evaluation. Each phase was implemented using open-source resources and free cloud compute.

3.1. Datasets

The human-written data was derived from the publicly available Wikipedia 20220301 English dataset on Hugging Face Datasets. 12000 paragraphs were sampled to ensure diversity in topic, style, and structure.

Data cleaning included:

- Filtering out overly short (less than 10 tokens) or excessively long (more than 500 tokens) segments.
- Removing non-English or low-quality entries (e.g., HTML tags, tables, or list items).
- Deduplicating near-identical text using MinHash to ensure dataset diversity.

The resulting dataset was split into train and validation sets, `human_train` (10800 paragraphs) and `human_val` (1200 paragraphs).

Each human paragraph was rewritten using two open-source models, Google Flan T5 Large and Gemma 2B, to create two synthetic datasets.

The generation prompt encouraged clarity and information density:

“Fully rewrite the following paragraph so that it is more informative, cohesive, and well-structured. Preserve factual accuracy and meaning while improving readability and conciseness.”

Synthetic generation was performed automatically in batches using the Hugging Face transformers pipeline, running on a Colab GPU. Quality control included:

- Filtering out empty or malformed generations.
- Rejecting overly short or repetitive outputs to maintain linguistic diversity.
- Conducting random manual checks to ensure factual alignment with the source text.

This produced the synthetic dataset (`synthetic_train.txt`) of comparable size and structure to the human corpus.

A fourth dataset was created by mixing the synthetic dataset created by Gemma 2B and the human-written dataset.

3.2. Model Selection and Training

A small open-source language model was used for continued pretraining to ensure feasibility under limited compute resources. The primary model selected for this study

was `Pythia-70M`, consisting of approximately 70 million parameters. This architecture was chosen because it is lightweight, reproducible, and well-documented within the Hugging Face model hub, enabling straightforward deployment on free-tier GPUs.

3.3. Evaluation and Metrics

Model performance was assessed through both intrinsic and extrinsic evaluation criteria. Intrinsically, language modeling quality was measured using validation perplexity and loss trends, reflecting each model’s ability to generalize and predict natural text sequences. Extrinsically, evaluation focused on factual accuracy and reasoning ability using small, representative benchmarks such as ARC-Easy. These metrics allow for a controlled assessment of how data source, human versus synthetic, influences a model’s linguistic coherence, knowledge retention, and reasoning generalization. Together, these metrics provide a balanced view of both statistical learning efficiency and task-level performance.

4. Experiments

4.1. Model Training

All models were trained using the causal language modeling (CLM) objective, where the model learns to predict the next token given previous context. Training was performed separately for the datasets under identical configurations to isolate the effect of data source. The overall setup is summarized below:

- **Optimizer:** AdamW
- **Learning Rate:** $1e-4$
- **Sequence Length:** 512 tokens
- **Batch Size:** 8 (with gradient accumulation of 4)
- **Training Steps:** 2,000–5,000
- **Precision:** Mixed precision (FP16)
- **Evaluation Frequency:** Every 200 steps

Four identical training runs were conducted:

1. **Run A (Human):** Model trained on the human-written Wikipedia subset.
2. **Run B (Synthetic-Flan T5):** Model trained on the Flan T5 Large rewritten dataset.
3. **Run C (Synthetic- Gemma 2B):** Model trained on the Gemma 2B rewritten dataset.
4. **Run D (Hybrid):** Model trained on the hybrid dataset with both human-written and synthetic data.

All other hyperparameters, random seeds, and initialization weights were kept constant to ensure a controlled comparison. Model checkpoints and logs were saved for later evaluation and reproducibility. The training process was executed using the Hugging Face `Transformers` library with PyTorch backend and managed on Google Colab equipped with NVIDIA A100.

4.2. Model Evaluation

I compared:

- **Perplexity** on a held-out Wikipedia validation set.
- **QA accuracy** on small subsets of ARC-Easy and HellaSwag to assess factual and reasoning generalization.
- **Qualitative analysis:** randomly sample model outputs to inspect fluency and factual correctness.

4.3. Computational Resources

- **Hardware:** Google Colab Pro (T4/A100 GPU).
- **Compute Time:** Approximately 5-6 days total for all runs.
- **Memory:** 16–24 GB VRAM.
- **Software:** PyTorch, Hugging Face Transformers, Datasets, and Accelerate. All components are free and open source.

5. Results

I evaluated Pythia-70M models fine-tuned under four training regimes: (1) human-written Wikipedia text, (2) synthetic text generated using Flan-T5-Large, (3) synthetic text generated using Gemma-2B, and (4) a mixed dataset consisting of 20% human-written data and 80% Gemma-generated synthetic data. All models are fine-tuned from the same pretrained checkpoint using identical hyperparameters (learning rate $1e-4$, 2 epochs), ensuring that observed differences are attributable to the training data.

5.1. Language Modeling Performance

Table 1 reports evaluation loss and perplexity on a held-out Wikipedia validation set. The model trained exclusively on human-written data achieves the lowest perplexity (60.42), indicating superior alignment with natural language statistics. Both synthetic-only models exhibit higher perplexity, with the Gemma-generated dataset producing the highest perplexity among single-source datasets. The Flan-T5 synthetic dataset results in intermediate performance.

Introducing a small proportion of human-written data into the Gemma-generated synthetic dataset substantially improves perplexity, reducing it from 81.95 to 69.60. This mixed-data configuration closes much of the gap between synthetic-only and human-only training, demonstrating that even limited amounts of human data can stabilize language modeling performance.

5.2. Reasoning Performance on ARC-Easy

Table 2 reports accuracy on a 50-question subset of ARC-Easy. While absolute accuracy remains low across all models due to the small parameter count, clear relative differences emerge. The Gemma-generated synthetic dataset achieves substantially higher accuracy (26%) than both the human-only model (4%) and the Flan-T5 synthetic model

Training Data	Eval Loss	Perplexity
Human (Wikipedia)	4.1014	60.42
Synthetic (Flan-T5)	4.2705	71.56
Synthetic (Gemma-2B)	4.4061	81.95
20% Human / 80% Gemma	4.2351	69.60

Table 1. Language modeling performance on a held-out Wikipedia validation set.

Training Data	Accuracy
Human (Wikipedia)	0.04
Synthetic (Flan-T5)	0.08
Synthetic (Gemma-2B)	0.26
20% Human / 80% Gemma	0.26

Table 2. ARC-Easy accuracy on a 50-question subset.

Training Data	Accuracy
Human (Wikipedia)	0.10
Synthetic (Flan-T5)	0.10
Synthetic (Gemma-2B)	0.12
20% Human / 80% Gemma	0.18

Table 3. HellaSwag accuracy on a 50-example subset.

(8%). Notably, the mixed dataset preserves the same ARC-Easy accuracy as the Gemma-only synthetic model, indicating that incorporating human-written data does not diminish the reasoning-related gains introduced by synthetic data.

5.3. Commonsense Reasoning on HellaSwag

I further evaluate commonsense reasoning using a 50-example subset of the HellaSwag benchmark, which requires selecting the most plausible continuation of a given real-world situation. Table 3 reports accuracy for each training regime.

Overall accuracy remains low across all models, reflecting the difficulty of commonsense reasoning for small language models. However, the mixed dataset achieves the highest accuracy, outperforming both human-only and synthetic-only training. This trend is consistent with results on ARC-Easy and suggests that combining a modest amount of human-written data with synthetic data can improve reasoning performance even when absolute performance remains limited.

5.4. Qualitative Analysis

To better understand differences in generation behavior beyond quantitative metrics, I qualitatively analyze model outputs on a Wikipedia paragraph continuation task. Each model is prompted with the same paragraph prefix and asked to generate a short continuation. Representative examples are shown below.

Example 1. *Context:* He was born in Aschaffenburg, Bavaria and educated in medicine at the universities of Munich, Berlin, and Strasbourg, where he received his doctorate in 1908. During the following year, he began clinical work under the psychiatrist Emil Kraepelin and did laboratory work with Franz Nissl and Alois Alzheimer in Munich.

Human-trained continuation: He was also a doctoral student at the University of Munich, where he studied medicine and applied for a doctorate in medicine. He was also a doctoral student at the University of Munich, where he studied medicine in Munich.

Synthetic-trained (Gemma) continuation: The following year, Kraepelin and his colleagues became central figures in early Alzheimer’s disease research, helping establish the field of neuropathology and solidifying Munich as a leading research center.

Synthetic-trained (Flan-T5) continuation: He was also a student at the University of Munich where he studied medicine. He was also a student at the University of Munich where he studied medicine.

In this example, the human-trained and Flan-T5 models exhibit strong repetition and limited semantic progression. In contrast, the Gemma-trained model produces a more fluent and varied continuation, though it introduces unsupported factual claims.

Example 2. *Context:* Jakob was the author of five monographs and nearly 80 scientific papers. His neuropathological research contributed greatly to the delineation of several diseases, including multiple sclerosis and Friedreich’s ataxia.

Human-trained continuation: He also described the disease as a disease of the nervous system. He also described the disease as a disease of the nervous system.

Synthetic-trained (Gemma) continuation: His work helped clarify the pathological mechanisms underlying these disorders and influenced subsequent research in neurology and clinical diagnosis.

Synthetic-trained (Flan-T5) continuation: His neuropathological research contributed greatly to the delineation of several diseases, including multiple sclerosis and Friedreich’s ataxia.

Here again, the Gemma-trained model generates a more coherent and explanatory continuation, while the human-trained and Flan-T5 models largely repeat content from the input.

Across examples, models trained on human-written data frequently exhibit degenerative repetition, while Flan-T5 synthetic data leads to paraphrastic but shallow continuations. Gemma-generated synthetic data produces more diverse and fluent text, but at the cost of increased hallucination and factual inconsistency. The mixed-data model behaves similarly to the Gemma-only model, suggesting that

synthetic data strongly shapes generation style even when human-written data is partially reintroduced.

6. Discussion

The results reveal a clear trade-off between language modeling fidelity and reasoning performance when training small language models on synthetic data. Human-written Wikipedia text yields the lowest perplexity, reflecting strong alignment with the Pythia model’s pretraining distribution. This outcome is expected, as Pythia-70M is pretrained primarily on large-scale human-written corpora, making human-written validation data inherently in-distribution.

In contrast, synthetic datasets introduce distributional shifts in phrasing, structure, and discourse patterns. Gemma-generated synthetic text is more elaborative and stylistically diverse, which appears to benefit reasoning tasks such as ARC-Easy, but negatively impacts perplexity when evaluated on human-written validation data. This mismatch suggests that perplexity alone may underestimate the utility of synthetic data, particularly for tasks that benefit from explicit reasoning structure.

The quality of synthetic data plays a critical role in downstream performance. Flan-T5-generated text largely preserves surface-level structure and phrasing, resulting in synthetic data that is only marginally different from the original corpus. Consequently, Flan-T5 synthetic data yields modest improvements over human data in reasoning tasks, but does not substantially alter model behavior. In contrast, Gemma-generated synthetic text introduces greater structural variation and explanatory content, leading to significantly improved reasoning accuracy despite higher perplexity. These findings confirm that the synthetic data generator plays a major role in the quality of data. Gemma 2B is a stronger and larger model, and it produces better quality data, which leads to better results in terms of reasoning.

The mixed-data experiment highlights the complementary roles of human and synthetic data. Incorporating only 20% human-written text into Gemma-generated synthetic data substantially reduces perplexity while preserving reasoning gains. This finding aligns with prior work on hybrid training strategies, which suggests that human data anchors linguistic realism while synthetic data expands coverage and reasoning diversity. Importantly, this result demonstrates that synthetic data need not fully replace human-written data to be effective; instead, strategic mixing can yield better overall performance.

Finally, the observed discrepancy between lower training loss and higher evaluation perplexity for synthetic datasets suggests overfitting to synthetic regularities rather than true generalization. This emphasizes the need for multi-faceted evaluation strategies when studying synthetic data, as training loss alone can be misleading.

6.1. Implications for Small-Model Training

These findings have important implications for training small language models under constrained compute and data budgets. While synthetic data may not perfectly match human language distributions, it can encode reasoning-relevant patterns that are difficult to extract from natural text alone. For small models, which are limited in capacity, such structured signals may be particularly valuable. Hybrid training regimes therefore, offer a practical pathway for improving reasoning performance without sacrificing language modeling quality.

7. Limitations

This study is subject to certain limitations. All experiments are conducted using a single small model (Pythia-70M), and results may not generalize directly to larger or instruction-tuned architectures.

Additionally, synthetic data quality is highly dependent on the choice of generator model and prompt design. Different prompting strategies or stronger generator models could produce substantially different outcomes.

Finally, evaluation is performed exclusively on Wikipedia-style text, which may disadvantage synthetic datasets when evaluated using perplexity and limits conclusions about cross-domain generalization.

8. Conclusion

This work presents an empirical comparison of human-written and synthetic data for training small language models under constrained compute budgets. It shows that human-written data consistently yields superior language modeling performance, while synthetic data, particularly text generated by Gemma-2B, substantially improves performance on reasoning tasks. Importantly, mixing a small proportion of human-written data with synthetic data significantly reduces the generalization gap without sacrificing reasoning gains.

Findings highlight that synthetic data quality and generator choice are critical, and that perplexity alone is insufficient to evaluate synthetic training regimes. Overall, this study demonstrates that carefully constructed hybrid datasets offer a promising path forward for efficient and effective small-model training.

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